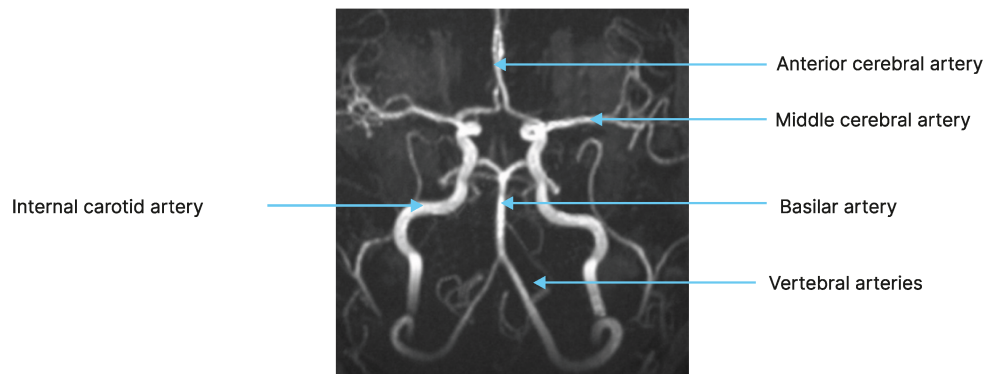
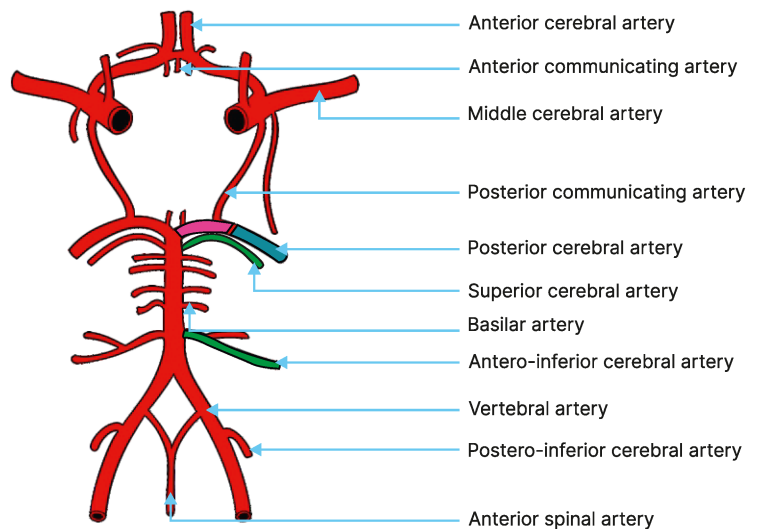




## BERRY ANEURYSM & POSTERIOR CEREBRAL ARTERY ANATOMY

- M/c **site of Berry aneurysm** → Anterior communicating artery
- M/c CN involved in Berry aneurysm → **Oculomotor nerve** (runs adjacent to posterior communicating artery)
- Posterior cerebral artery:
  - Pink part → P<sub>1</sub> segment
  - Blue part → P<sub>2</sub> segment



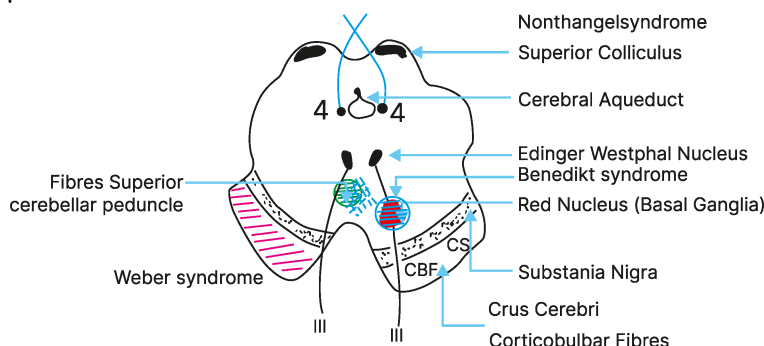
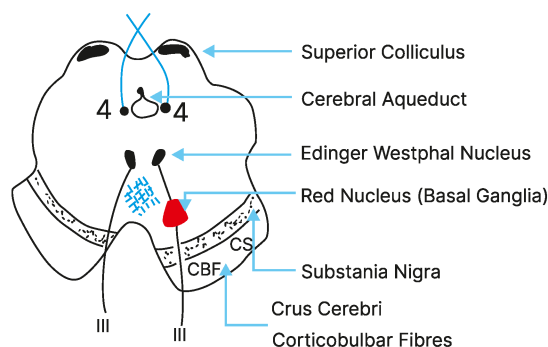
| LESION SITE                   | MANIFESTATION                                                                                                  |
|-------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>P<sub>1</sub>PCA</b>       | <ul style="list-style-type: none"> <li>• Midbrain stroke</li> </ul>                                            |
| <b>P<sub>2</sub>PCA</b>       | <ul style="list-style-type: none"> <li>• Occipital lobe- Homonymous hemianopia with macular sparing</li> </ul> |
| <b>Basilar artery</b>         | <ul style="list-style-type: none"> <li>• Pontine stroke</li> </ul>                                             |
| <b>Vertebral artery</b>       | <ul style="list-style-type: none"> <li>• Lateral medullary syndrome</li> </ul>                                 |
| <b>Anterior spinal artery</b> | <ul style="list-style-type: none"> <li>• Medial medullary syndrome</li> </ul>                                  |

### Important Information

- Middle cerebral artery → Not a part of Circle of Willis

## ANATOMY KEY STRUCTURES AND PATHWAYS

- Superior colliculus → Plays a role in vertical gaze control
- Trochlear nerve → Only CN with dorsal origin and decussates before exiting
- Edinger-Westphal nucleus → Parasympathetic nucleus of the oculomotor nerve
- Red nucleus → Connects midbrain with basal ganglia
- Substantia nigra → A/w vascular Parkinsonism
- Crus cerebri → Corticospinal and corticobulbar tracts pass through; CBF supplies face



| SYNDROME                  | STRUCTURES INVOLVED                                                                                                                                                               | FEATURES                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Weber syndrome</b>     | <ul style="list-style-type: none"> <li>• 3<sup>rd</sup> nerve</li> <li>• Corticospinal fibres supplying arms &amp; legs</li> <li>• Corticobulbar fibres supplying face</li> </ul> | <ul style="list-style-type: none"> <li>• I/L Ptosis, Mydriasis, Eye looking down &amp; out</li> <li>• C/L Face weakness</li> <li>• C/L Arm and leg weakness</li> </ul>     |
| <b>Nothnagel syndrome</b> | <ul style="list-style-type: none"> <li>• 3<sup>rd</sup> nerve</li> <li>• Superior cerebellar peduncle decussation</li> </ul>                                                      | <ul style="list-style-type: none"> <li>• I/L 3<sup>rd</sup> nerve palsy</li> <li>• C/L gait ataxia</li> </ul>                                                              |
| <b>Benedikt syndrome</b>  | <ul style="list-style-type: none"> <li>• 3<sup>rd</sup> nerve</li> <li>• Rubrothalamic tract</li> <li>• Medial lemniscus</li> </ul>                                               | <ul style="list-style-type: none"> <li>• I/L 3<sup>rd</sup> nerve palsy</li> <li>• C/L Chorea</li> <li>• C/L Resting tremors</li> <li>• C/L Proprioception loss</li> </ul> |



Ptosis

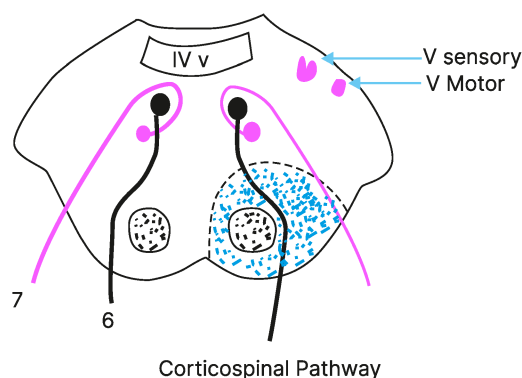
## BRAINSTEM STROKE FEATURES & COMMON ETIOLOGY

- Involvement of brain stem manifests with crossed features
- M/c cause of acute ischemic stroke → Non-valvular atrial fibrillation

- **Locked-in syndrome** → Features of both **Nothnagel** and **Benedikt** syndromes due to injury to both red nucleus and superior cerebellar peduncle
- **Parinaud syndrome** → **Pinealoma** → impaired upward gaze + sun-setting sign
- **Thalamic Dejerine-Roussy syndrome**:
  - Occlusion of penetrating branches of Thalamic & Thalamogeniculate arteries
  - Hemisensory loss → followed by agonizing searing, burning pain

## PONTINE STROKE

01:16:43



| FOVILLE SYNDROME                                                                                                                                                                                                                                                                                                      | MILLARD GUBLER SYNDROME                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Dorsal pontine injury involves 6<sup>th</sup> nerve nucleus</li> <li>• 6<sup>th</sup> nerve nucleus communicates with C/L 3rd nerve nucleus → simultaneous lateral rectus and medial rectus movement of opposite eyes → injury prevents coordinated muscle action</li> </ul> | <ul style="list-style-type: none"> <li>• Ventral pontine (6<sup>th</sup>, 7<sup>th</sup> &amp; corticospinal pathway) injury</li> </ul>      |
| <ul style="list-style-type: none"> <li>• I/L Lateral gaze palsy</li> <li>• I/L Facial nerve palsy</li> <li>• C/L Hemiparesis</li> </ul>                                                                                                                                                                               | <ul style="list-style-type: none"> <li>• I/L Lateral rectus weakness</li> <li>• I/L Facial nerve palsy</li> <li>• C/L Hemiparesis</li> </ul> |

## LOCKED IN SYNDROME

01:23:53

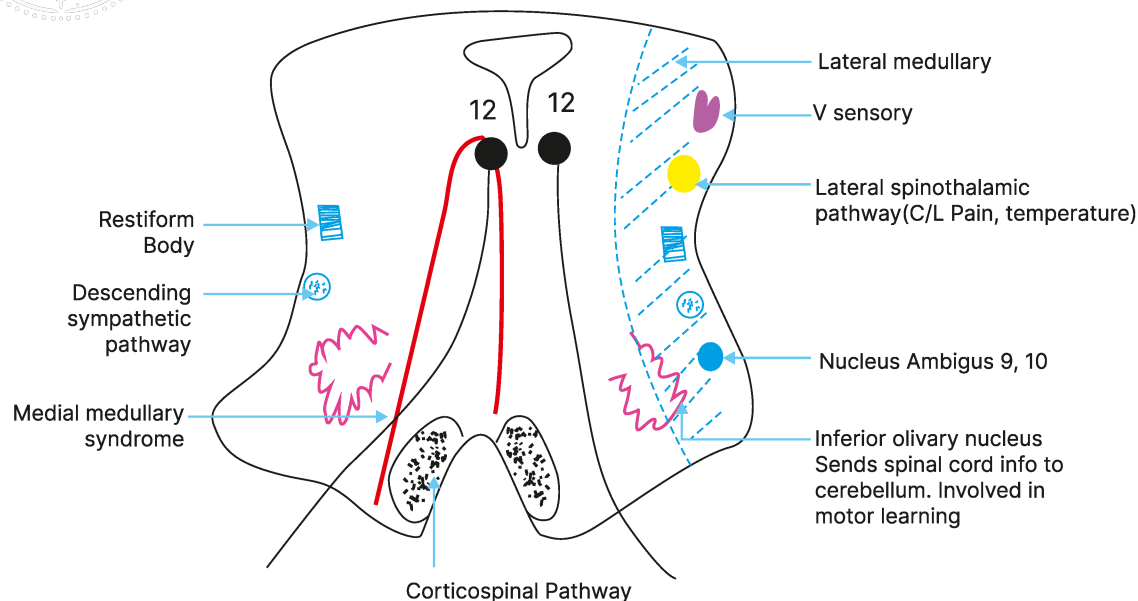
- Damage to Basilar artery and its branches

## MANIFESTATIONS

- B/L corticospinal tract injury → quadriplegia + B/L Babinski sign
- Can't speak, eat & drink due to involvement of CN 9, 10 & 12
- Preserved cognition → Intact reticular activating system & cortex
- Only vertical eye movement are possible (horizontal affected) due to damaged to PPRF in pons
- Rostral interstitial nucleus of the medial longitudinal fasciculus (riMLF) in the midbrain □ Primary centre for vertical gaze



## MEDULLARY ANATOMY & SYNDROMES: KEY STRUCTURES AND PATHWAYS

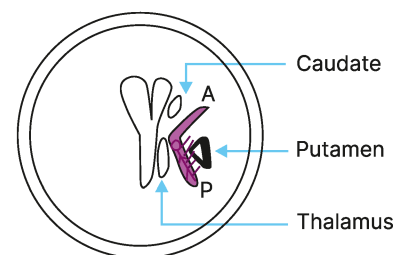


- Lateral spinothalamic pathway- Carries C/L Pain, and temperature
- Nucleus Ambigus- Motor feed to CN 9 & 10
- Descending sympathetic pathway- Manifestations of Horner syndrome
- Inferior olivary nucleus → Relays spinal cord info to cerebellum → Involved in motor learning

| MEDIAL MEDULLARY SYNDROME                                                                                                                                                                                                                   | LATERAL MEDULLARY SYNDROME                                                                                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Anterior spinal artery</li> <li>• I/L 12<sup>th</sup> nerve palsy</li> <li>• I/L tongue deviation on tongue protrusion. Remember as 'Tongue licks the lesion'</li> <li>• C/L Hemiplegia</li> </ul> | <ul style="list-style-type: none"> <li>• Vertebral artery &gt; PICA</li> <li>• Crossed Hemianesthesia</li> <li>• I/L hypotonia of arms and limb ataxia</li> <li>• I/L Horner syndrome</li> <li>• Dysphagia and dystonia</li> <li>• CN 5, 7, 8, 9, 10 &amp; 11 are also involved</li> </ul> |

## LACUNAR STROKE

- Small vessel stroke → Penetrating branches of lenticulostriate artery
- Lipohyalinosis of lenticulostriate artery branches supplying posterior limb of internal capsule → C/L face, arm & leg weakness (pure motor stroke)





- Q. A 65-year-old T2 DM man develops sudden onset face dropping and right sided hemiparesis while having breakfast. On examination in ER he is found to have left sided lateral gaze palsy and left facial weakness. Diagnosis is?
- a. Foville syndrome
  - b. Millard Gubler syndrome
  - c. Weber syndrome
  - d. Claude syndrome

Ans (a)

- Q. A 58-year-old presents with complaints of sudden onset vertigo, difficulty swallowing and slurred speech for past 6 hours. She says her left side of face feels numb and she cannot feel any sensation. On examination, hypotonia in arms is noted with left sided ptosis and miosis and right arm and leg sensory loss to pain and temperature. Which blood vessel is involved?
- a. Vertebral artery
  - b. PICA
  - c. P<sub>1</sub>PCA
  - d. P<sub>2</sub>PCA

Ans (a)

- Q. Which of the following is correct about Lacunar stroke?
- a. Contralateral face, arm and leg weakness
  - b. Contralateral face, arm weakness, aphasia, agraphia
  - c. Contralateral face, arm weakness with diplopia, dysarthria and dysphagia
  - d. Contralateral face, arm weakness with resting tremor and chorea

Ans (a)



# HEADACHE AND MIGRAINE

## HEADACHE AND MIGRAINE COMMON CAUSES OF HEADACHE

00:01:07

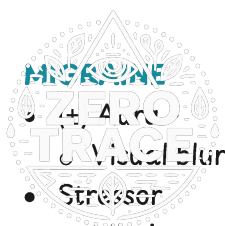
| CAUSES       | PRIMARY HEADACHE                                                                                                                                                                                    | SECONDARY HEADACHE                                                                                                                                  |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| M/c          | <ul style="list-style-type: none"><li>• <b>Tension headache</b><ul style="list-style-type: none"><li>○ &gt; 15 days in a month</li></ul></li></ul>                                                  | <ul style="list-style-type: none"><li>• Infection<ul style="list-style-type: none"><li>○ Meningitis, Sinusitis</li></ul></li></ul>                  |
| Least common | <ul style="list-style-type: none"><li>• <b>Cluster headache</b><ul style="list-style-type: none"><li>○ Deep seated retro orbital pain</li><li>○ Recurrence of pain at same time</li></ul></li></ul> | <ul style="list-style-type: none"><li>• <b>Brain tumor</b><ul style="list-style-type: none"><li>○ Vomiting followed by headache</li></ul></li></ul> |

### BUZZWORDS TO PICK UP DIAGNOSIS

- Worst headache of my life/ Thunderclap headache/ Headache peaking in 5 minutes with nuchal rigidity - **Subarachnoid hemorrhage**
- 70-year-old man with temporal headache + scalp tenderness/ pain on combing hair - **Giant cell arteritis**
  - Extracranial cause of headache, U/L
  - Along with fever of unknown origin
- Vomiting that precedes headache by couple of weeks - **↑ICP, Brain tumor**
  - Holocephalic headache
- Migraine patient taking NSAIDs + Caffeine + Opioids and still headache persists and occurs daily - **Medication over use headache**
  - **Opioids and Barbiturates** should **not be used** in T/t of migraine
    - Can cause habituation and withdrawal symptoms
- Positional headache/ increases when standing and decreases on sitting/ lying down - **Low CSF volume headache**
  - Causes - LP, Epidural procedure, weightlifting, excessive straining, coughing, multiple orgasm
  - It begins with 48 hours but can be delayed for up to 12 days
- Obese female presents with headache on awakening and increases on lying down, Fundus shows papilledema - **Idiopathic intracranial HTN/ Pseudotumor cerebri**
  - T/t - **Acetazolamide**
  - MRI and MR venogram - partial obstruction in venous sinuses (↓ reabsorption of CSF)
  - Guarded LP - ↑ ICP and removal of CSF causes improvement of headache

### Important Information

- Normal pressure hydrocephalus - CSF removal improves gait (**Fisher test**)

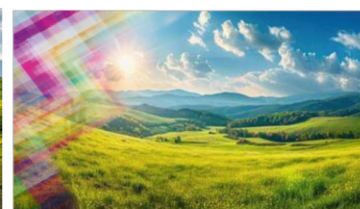


Visual blurring, Zigzag lines, Scotoma

- Stressor
  - Missing a meal, Sunlight
  - Physical exertion
- Female, pulsatile, throbbing nature of headache
  - Photophobia, phonophobia
  - Duration(onset) - 4 hours - 72 hours (average ~ 24 hours)
  - U/L
  - Nausea, vomiting
  - Disabling in character (↑ in severity with neck movement)
- 4 out of 5 symptoms, ≥ 5 attacks/ year → Migraine



Visual blurring



Zig-zag lines

### INTERNATIONAL HEADACHE SOCIETY - CRITERIA FOR MIGRAINE DIAGNOSIS

- Repeated attacks lasting 4-72 hours with normal physical examination and no other reasonable causes
- Normal physical examination -
  - No nuchal rigidity
  - No focal neurological deficits
  - No Cranial nerve palsy

| ANY 2 OF FOLLOWING                                                                                                                                             | ANY 1 OF FOLLOWING                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Moderate to severe intensity</li> <li>• Aggravation of movement</li> <li>• Throbbing nature</li> <li>• U/L</li> </ul> | <ul style="list-style-type: none"> <li>• Nausea/ vomiting</li> <li>• Photophobia</li> <li>• Phonophobia</li> </ul> |

### MIGRAINE DISABILITY ASSESSMENT TEST

#### MIGRAINE DISABILITY ASSESSMENT SCORE (MIDAS)

**Instructions:** Please answer the following questions about all headaches you have had in the last 3 months. Write your answer in the box next to each question. Write zero if you have not done the activity in the last 3 months

|   |                                                                                                                                                                                                                      |                                                |
|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| 1 | How many days in the last 3 months have you missed work or school because of your headaches?                                                                                                                         | <input type="text"/> <input type="text"/> Days |
| 2 | How many days in the last 3 months has your performance at work or school been reduced by half or more because of your headaches? (Do not include the days you counted in question 1 that you missed work or school) | <input type="text"/> <input type="text"/> Days |
| 3 | How many days in the last 3 months have you not done housework because of your headaches?                                                                                                                            | <input type="text"/> <input type="text"/> Days |

4 On how many days in the last 3 months has your housework performance been reduced by half or more due to your headaches? (Do not include the days you counted in question 3 when you did not do housework)   Days

5 How many days in the last 3 months have you missed family, social or leisure activities because of your headaches?   Days

**Total**   Days

A How many days in the last 3 months have you had headaches? (If **the headache lasted more than 1 day, count each day**)   Days

B **On a scale of 0 to 10, on average, how intense were these headaches?** (Where 0 = no pain, and 10 maximum possible pain)  Days

Scoring: After completing the questionnaire, add up the total number of days for questions 1-5. Do not include answers to questions A and B in the overall score

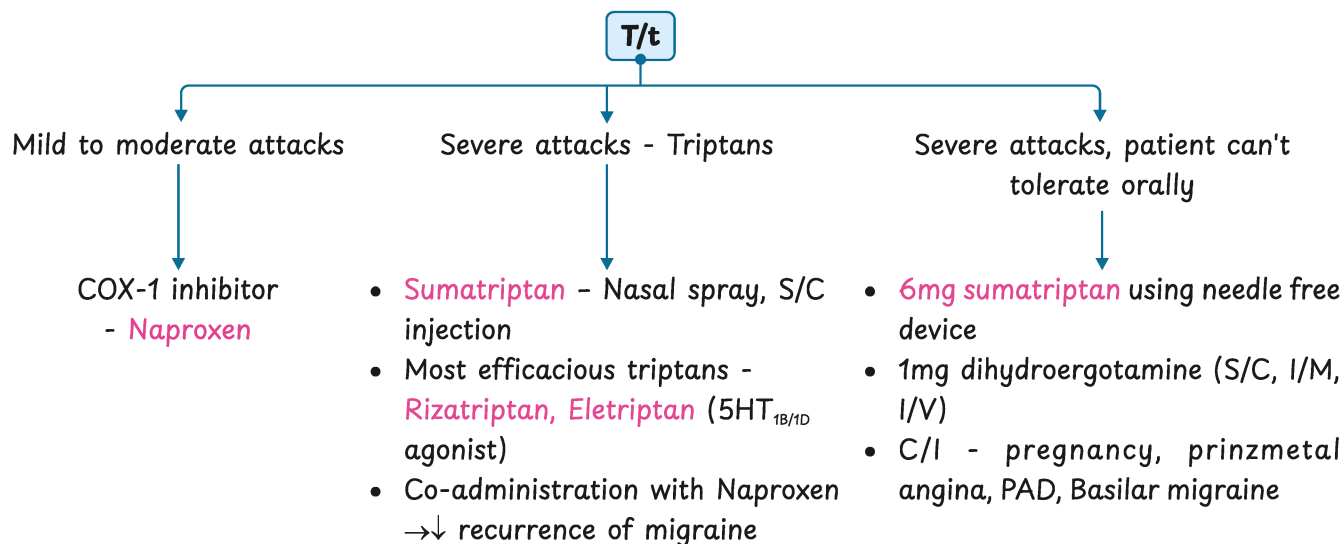
### GRADING SYSTEM FOR THE MIDAS QUESTIONNAIRE

| MIDAS DEGREE | DEFINITION               | MIDAS SCORE |
|--------------|--------------------------|-------------|
| I            | Minimal or no disability | 0-5         |
| II           | Mild disability          | 6-10        |
| III          | Moderate disability      | 11-20       |
| IV           | Severe disability        | >21         |

- Disability - MIDAS
- Score **> 21** → severe disability
  - Prophylaxis needed

### T/t FOR MIGRAINE

00:27:50



## OTHER APPROACHES IN MIGRAINE MANAGEMENT

- 5-HT<sub>1F</sub> Calcitonin Gene Related Peptide Receptor Antagonists
  - Frogepant
  - Ubrogepant
- 5HT<sub>1F</sub> receptor agonist - Ditans
  - Lasmiditan
  - No vascular effects → suitable for CVS and CNS disease in patients
  - Side effects - dizziness
  - Don't drive for 8 hours after tablet
- Non selective 5-HT receptor agonists
  - Ergotamine
- Single pulse transcranial magnetic stimulation (STMS) - Neuromodulation

## PREVENTION OF MIGRAINE

- Beta blockers - Propranolol
- Antidepressants - Amitriptyline, Venlafaxine
- Anticonvulsants - Topiramate, Valproate
- Others - Flunarizine, Candesartan
- CGRP antagonists
  - Galcanezumab
  - Erenumab
  - Eptinezumab - S/C 3 monthly basis
- Onabotulinum toxin injection in procerus, corrugator muscles
- STMS

## CASE SCENARIOS

1) A 25 year old female patient wakes with an excruciating headache at 5 am in morning. She takes a crocin tablet but headache has increased in severity. Subsequently she has multiple vomiting episodes. On examination she has ptosis in left eye with nuchal rigidity. First differential diagnosis is?

- a. Meningitis
- b. SAH
- c. ICSOL
- d. Migraine

Ans: (b)

Inference -

- Ptosis - presence of focal neurological deficit
  - Presence of nuchal rigidity
  - Short history → ICSOL → ruled out
  - Meningitis → should have fever/ case has short history → ruled out
- } Rule out -Migraine

## SAH

- NCCT head → blood in Sylvian fissure, **star of death appearance**
- Next best investigation after NCCT head in SAH → **cerebral angiography**
- TOC → SAH → **endovascular coiling**

- DCC for SAH - **Nimodipine** (to prevent primary vasospasm)
- Leading cause of SAH - trauma > rupture of berry aneurysm
- Location of berry aneurysm - **junction** of Anterior communicating artery and Anterior cerebral artery
- M/c CN involved in PCOM berry aneurysm - **CN III palsy**
- Risk of rupture (directly related to size) location - **Top of Basilar artery**

2) Young female while watching a movie in cinema hall started having severe retro-orbital pain and multiple episodes of vomiting. On examination redness is noted in right eye with vertically dilated pupil with sluggish reaction to light. First differential diagnosis is?

- Migraine with aura
- Glaucoma
- SAH
- Cluster headache

Ans: **(b)**

Inference

- Migraine with aura
  - Cluster headache
  - SAH → ruled out → absence of nuchal rigidity
- } Ruled out d/t presence of vertically dilated pupil

### Important Information

Angle closure glaucoma

- Most cases triggered in dark (E.g. movie theatre)
- Mydriasis → ciliary block → ↑ IOP → glaucoma
- T/t - Mannitol

## FAMILIAL HEMIPLEGIC MIGRAINE

00:46:23

- Features of migraine with weakness
- Mimics like stroke symptoms but weakness resolves gradually
- Calcium channel defect

### Important Information

Calcium channel defects in neurology

- Lambert Eaton syndrome (Oat cell carcinoma of lung)
- Hypokalemic periodic paralysis type I
- Familial hemiplegic migraine

## CASE SCENARIO

A 70 year old retired army man presents with feeling feverish for past 4 weeks for which he takes PCM on and off. He complains of scalp tenderness while combing hair for the same duration and pain in jaw while eating food. Dental examination is normal. Cord like structure is felt at right temporomandibular joint. Work up shows ESR-100 mm fall in 1<sup>st</sup> hour. Probable diagnosis?

Ans: Giant cell arteritis



- Aribi, temporal artery involved (luminal narrowing of blood vessel)
  - Supplies skin of scalp
  - Supplies TMJ
- } Vasculitis → ischemia → scalp tenderness/ jaw claudication
- Cord like structure → thrombosed auriculotemporal artery
  - M/c clinical presentation - **fever of unknown origin**
  - Ophthalmic artery involvement → monocular blindness

### WORKUP

- Biopsy of thrombosed vessel
  - Giant cells
  - Granulomatous vasculitis
- ESR↑

### T/t

- Steroids
- No role of triptans

### Important Information

ESR - 100 mm fall in first hour

- Subacute bacterial endocarditis
- Giant cell arteritis
- Multiple myeloma

### CASE SCENARIO

40 year old lady during consultation for headache says, "I have two children who are phone addicts, and their academic performances has dropped, and I'm very stressed about it." She describes pain as if someone has tied a piece of tight constrictive band around her head. BP is normal. Refraction testing shows visual acuity of 6/6. Sinus/ Fundus examination is normal. You asked her to keep pain diary and it shows 20 days of headache per month.

### INFERENCE

- Recurrence of headache 20 times/ month
  - Character of headache - band tied around head
  - No hypertension
  - Normal refraction
  - Normal fundus findings
  - Normal sinus
- } Ruled out: ↑ ICP, refraction errors, sinusitis

**DIAGNOSIS** - Tension type headache

### T/t

- Paracetamol, Amitriptyline

### CASE SCENARIO

25 year old guy presents with deep seated retro-orbital pain in right eye. He also tells that attacks occur daily



at same time of day for few weeks of each year and are triggered with alcohol intake. On examination nasal congestion is noted in right eye along with nasal stuffiness.

### INFERENCE

- Deep seated retro orbital pain
- Recurrence of pain at same time
- Alcohol → trigger
- Epiphora, ptosis, lacrimation, nasal stuffiness
- Pacing, restlessness

**DIAGNOSIS** - Cluster headache (psychiatric manifestation make this condition being called as 'suicide headache')

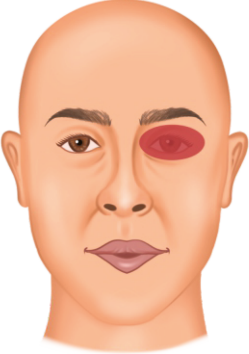
- No. of attacks/ day → 1-8/day
- Duration of attacks → 15 min - 180 min

### T/t

- 1<sup>st</sup> step - High flow O<sub>2</sub> at 12-15 L/min
- Sumatriptan (s/c) (abortive t/t) (DOC)
- Lithium

### PREVENTION

- **Verapamil** – best option
- Greater occipital nerve blocks
- Noninvasive vagal nerve stimulation (nVNS)
- Transcranial magnetic stimulation
- Steroids
- Melatonin

| CLUSTER HEADACHE                                                                    | MNEMONIC TO REMEMBER FEATURES                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <p><b>"SP's of Cluster Headache"</b></p> <p><b>P: Pain</b> (severe, unilateral, orbital/periorbital and called Suicide headache)</p> <p><b>P: Periodic</b> (clustered attacks, often same time daily)</p> <p><b>P: Ptosis</b> (Partial Horner's syndrome)</p> <p><b>P: Pacing</b> (restlessness during attacks)</p> <p><b>P: Parasympathetic symptoms</b> (tearing, nasal congestion)/Autonomic features</p> <p>Durations of attack: 15 minutes to 3 hours</p> <p><b>Management of cluster headache: "SO FAST"</b></p> <p><b>1. S: Sumatriptan</b> (Subcutaneous or intranasal for acute attack)</p> <p><b>2. O: Oxygen therapy</b> (100% oxygen via non-rebreather mask)</p> <p><b>3. F: First-line prophylaxis</b> (Verapamil)</p> <p><b>4. A: Avoid triggers</b> (alcohol, stress, etc.)</p> <p><b>5. S: Steroids</b> (Short course of prednisone for cluster periods)</p> <p><b>T: Transitional therapy</b> (e.g., bridging with steroids or dihydroergotamine)</p> |



- Neuralgiform headache
- Conjunctival injection and tearing
- Character - Burning/ stabbing periorbital pain
- 3 - 200 attacks/ day
- Duration of each attack - 5 - 240 seconds

### CLUSTER HEADACHE VS SUNCT

| FEATURES            | CLUSTER HEADACHE                                                                                  | SUNCT                                                                                 |
|---------------------|---------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| NO. OF ATTACKS/ DAY | 1 - 8                                                                                             | 3 - 200                                                                               |
| DURATION OF ATTACKS | 15 min - 180 min                                                                                  | 5 - 240 sec                                                                           |
| ABORTIVE T/t        | Sumatriptan injection/nasal spray                                                                 | I/V Lidocaine                                                                         |
| PREVENTIVE T/t      | <ul style="list-style-type: none"> <li>• Verapamil, nVNS</li> <li>• Melatonin, Lithium</li> </ul> | <ul style="list-style-type: none"> <li>• Lamotrigine</li> <li>• Topiramate</li> </ul> |

### HEADACHE FEATURES THAT SUGGEST SERIOUS UNDERLYING DISORDER

- Sudden severe headache
- First severe headache
- Worst headache ever
- Vomiting preceding headache
- Sub-acute worsening over days and weeks
- Pain induced while bending, lifting and coughing - low CSF volume headache
- Pain that disturbs sleep - Cluster headache
- Pain that is present on awakening - HTN, OSA, IIH
- Known systemic illness
- Onset after 55 years - stroke/ brain tumor
- Fever/systemic symptoms - GCA
- Abnormal neurological examination
- Pain with local tenderness in area of temporal artery



Q. Which of the following case scenarios of headache warrants further investigation?

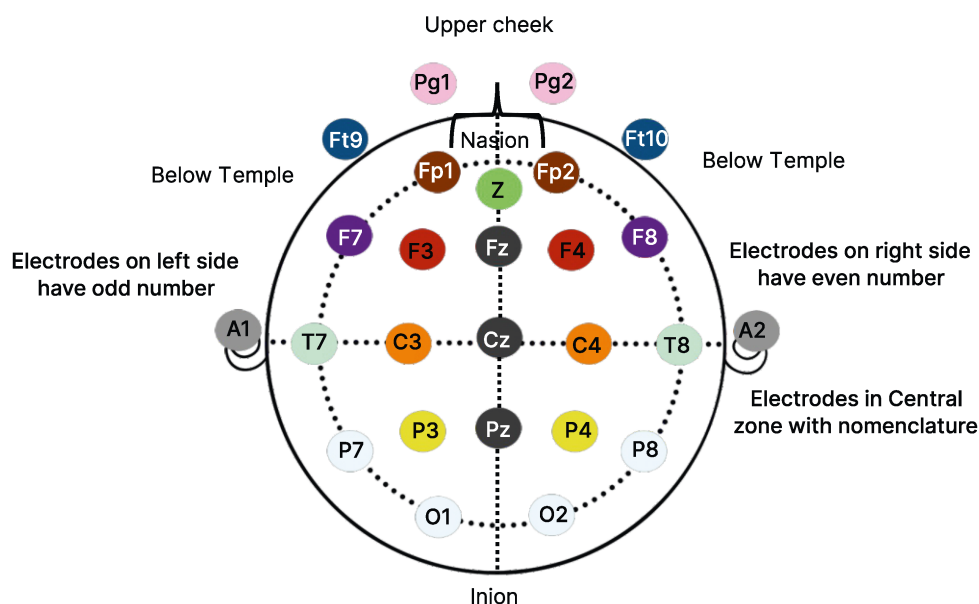
- a. 50 year old man with gradual increase of headache and vomiting
- b. 70 year old man with scalp tenderness
- c. Weightlifter with daily headache increasing on coughing and bending
- d. Daily headache in a woman with marital problems

- 1. a,b
- 2. a,b,c
- 3. a,b,c,d
- 4. a,c,d

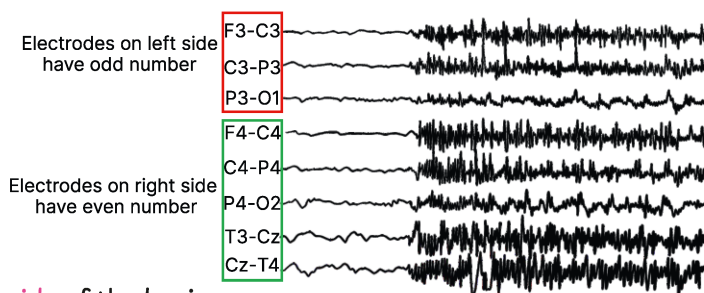
Ans: (2)

## INTRODUCTION

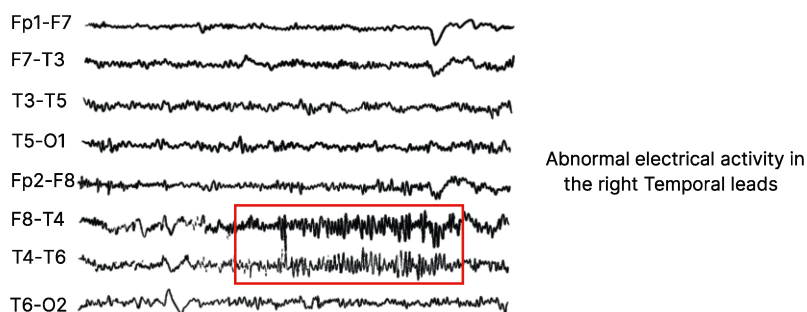
00:01:19



- Electrodes in midline subscript 'z'
- Electrodes on **left side** : denoted with **odd number**
- Electrodes on **right side** : denoted with **even number**

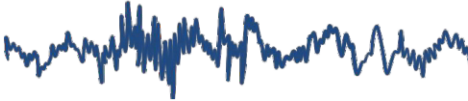


- **Superior leads** : For **left side** of the brain
- **Inferior leads** : For **right side** of the brain
- GTCS is depicted in the EEG



- Temporal lobe epilepsy / Focal seizures

- EEG recording is called **Montage**
- EEG invented by **Hans Berger**

| SITUATION               | TYPE OF WAVES                | EEG WAVE                                                                                                       | FREQUENCY |
|-------------------------|------------------------------|----------------------------------------------------------------------------------------------------------------|-----------|
| Eyes closed and relaxed | $\alpha$                     |                              | 8-12 Hz   |
| Awake                   | $\beta$                      |                              | 13-30 Hz  |
| NREM stage-1            | Theta                        |                              | 4-7 Hz    |
| NREM stage-2            | Sleep spindle<br>K-complexes |                              |           |
| NREM stage-3            | Delta waves                  | <br>Slowest, high amplitude | 0.5-4 Hz  |
| REM<br>(tone ↓)         | Sawtooth waves               |                            |           |

## Important Information

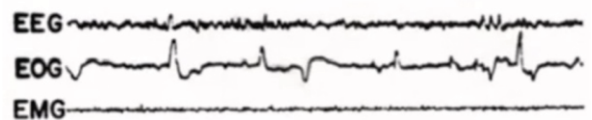
Delta waves in ECG is seen in **WPW syndrome** (Accessory pathway mediated tachycardia)

Saw tooth waves in ECG Atrial flutter

## Analyze the pattern of EEG:

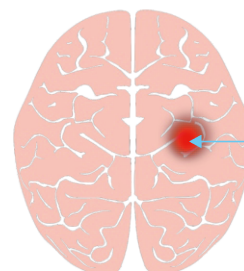
- EOG- rapid eye movement → spikes
- EMG touches the baseline → muscle tone ↓
- EEG - relatively small, pointed waves

} REM sleep

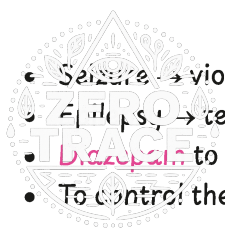


## SEIZURE vs CONVULSION vs EPILEPSY

- 5-10% population will have at least one seizure with highest incidence in childhood
- EEG can be normal in patients with epilepsy
  - Deep seated epileptogenic focus → normal EEG
- EEG is not mandatory for diagnosis of epilepsy
- Epileptogenic focus → abnormal electrical discharge → seizure



Abnormal focus in the brain producing abnormal electrical discharge



- Seizure → violent jerking → Convulsion
- Epilepsy → tendency to have recurrent seizures
- **Diazepam** to control convulsion
- To control the seizure → **anti-seizure/ anti-epileptogenic drug**

## GTCS - GENERALIZED TONIC-CLONIC SEIZURE

00:17:58

- Premonitory symptoms
    - Irritable, tired, malaise
    - Headache, dizziness
  - Tonus – loss of consciousness with ictal cry
    - 10-20 secs
    - Up-rolling eyes
    - Pooling of secretions/ frothing corners
    - Perioral cyanosis
    - (±) tongue bite
  - Clonus : Violent jerking of all limbs
  - Post ictal period- unconscious, noisy breathing, urine/ stool incontinence
    - Gradual orientation
  - Clonus T/t in hospital: I/V or I/M Lorazepam/ Rectal Diazepam
  - GTCS: **DOC - Sodium valproate**
- 30-60 sec
- Aspiration  
Temp ↑  
Metabolic acidosis

## EPILEPSY

00:27:29

- ≥ 2 episodes of unprovoked seizures
  - Seizure: abnormal electrical discharge may or may not get recorded
- Motor manifestation of seizure → : Convulsions
- Non motor manifestation → complex hallucinations
  - Smell of rotten fish
  - Micropsia/ Macropsia

## GCSE - GENERALIZED CONVULSIVE STATUS EPILEPTICUS

- Known GTCS patient with non-compliance to medication having seizures
  - **Duration > 5 mins**
- Impending and early SE- (**5 - 30 min**)
  - I/V **Lorazepam/ Midazolam /Clonazepam**
- I/V infusion phenytoin/ valproate / levetiracetam
- Established and early refractory SE (**30 min - 48 hours**)
  - I/V **Midazolam (dose ++)**
  - Propofol
- Late refractory SE (**> 48 hours**)
  - Medically induced coma
  - **Pentobarbital coma**
  - Under general anesthesia

## CLASSIFICATION OF SEIZURES

00:37:34

### FOCAL SEIZURE

- Twitching of mouth or involuntary movement in one limb
- Presence of Aura: (visual complaints)
- Focal seizure with intact awareness
  - History given by patient
- Focal seizure with impaired awareness
  - History given by others
  - Motor manifestations: face, upper limb jerky movements
  - Non-motor manifestations: burning kerosene/ rubber
- Focal seizure with secondary generalization
  - GTCS

### GENERALIZED ONSET SEIZURES

00:41:03

- Motor manifestations
  - Tonus clonus
  - Atonic seizures (drop attack)
  - Myoclonic seizures (sudden jerks)
- Non-motor manifestations
  - Absent seizures (vacant stares)
  - Atypical seizures

### FOCAL SEIZURES

00:48:58

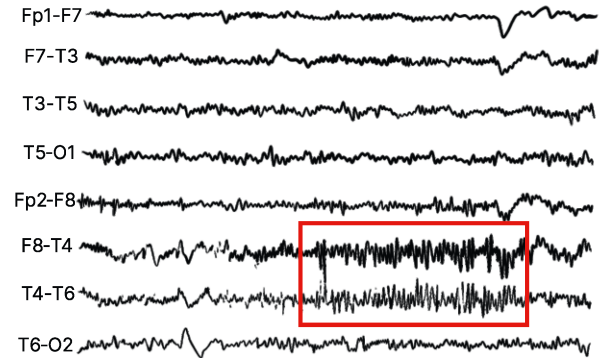
- Involuntary movements in a specific area/ part
- Causes
  - Neurocysticercosis
  - Mesial temporal lobe epilepsy (hippocampal sclerosis)

### FOCAL SEIZURE WITH INTACT AWARENESS

- Involuntary movements history given by the patient
  - Jacksonian march (Marching epilepsy)
    - Started at distal region and moving proximally
  - Todd's palsy
    - Weakness at the end of the seizure
    - Lasts from hours to days
  - Epilepsia partialis continua
    - Seizure persists from minutes to hours to days
  - Vertigo, paresthesias
  - Sensation of burning rubber/ kerosene
  - Subjective internal events
  - Micropsia
  - Macropsia
- } Subjective internal events

## FOCAL SEIZURES WITH IMPAIRED AWARENESS

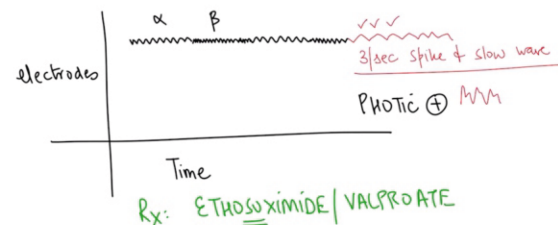
- A m. 1 motionless stare
- Autonomic
  - Repeated lip-smacking
  - Repeated swallowing
  - Repeated picking movement
  - Repeated chewing
  - Uncontrollable laughter/ crying
- Post ictal confusion
- Anterograde amnesia
- Transient deficits: Aphasia
- EEG: abnormal electrical activity in temporal lobe



Focal seizures

## GENERALIZED ONSET OF SEIZURES

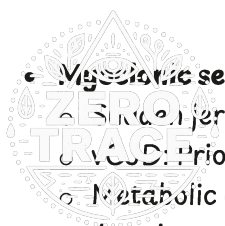
- **Typical absence seizures**
  - 4-10 years old
  - Loss of consciousness: 1-2 secs
  - Postural control maintained
  - Vacant staring spells/ day dreamer
  - Repeated blinking of eyes
  - Triggers: photic stimulation, hyperventilation
  - Post ictal deficit: absent
  - EEG findings:
    - With closed eyes:  $\alpha$  waves
    - With open eyes:  $\beta$  waves
    - With photic stimulation: 3/sec spike and slow wave pattern
  - T/t: Ethosuximide (young children)/ Valproate (older children)
  - Severity ↓ with ↑ in age
- **Atypical absence seizures**
  - EEG: < 2.5 Hz spike and slow wave pattern
  - Lapse of consciousness: longer duration
  - A/w mental retardation/ Lennox Gastaut Syndrome
- **Atonic seizures**
  - Loss of postural control for few seconds → injury prone
  - Loss of consciousness for few seconds
  - Repeated head nodding/ drop attacks
  - Post ictal deficit: Absent
  - Transient presence of (+) Babinski sign and (+) corneal reflex



00:53:07

### Important Information

Post ictal deficit absent in:  
 Absence seizure  
 Atonic seizure  
 Febrile seizure



## • Myoclonic seizures

- Sudden jerky movements
- CJD: Prion disease
- Metabolic encephalopathy
- Anoxic cerebral injury
- HACE (Hypobaric hypoxia)
- EEG: **Periodic sharp wave complexes**

## • Epileptic spasms

- Infant: myoclonic jerks involving the trunks → **salaam seizure**
- EEG: **Hypsarrythmia (grossly chaotic pattern)**
- BG rhythm suppression: Electro decremental response

Myasthenia gravis:  
Repetitive nerve test  
Decremental response

## • GTCS

### HOW TO DIFFERENTIATE UNCONSCIOUSNESS DUE TO VASOVAGAL SYNCOPE vs GTCS

| SYMPTOMS                     | VASOVAGAL SYNCOPE                                                         | GTCS            |
|------------------------------|---------------------------------------------------------------------------|-----------------|
| <b>Trigger</b>               | (+)<br>E.g. sunlight, blood, hospital setting, pain, emotional trauma etc | None            |
| <b>Tonus - clonus</b>        | < 15 sec                                                                  | 30 - 60 sec     |
| <b>Tongue bite</b>           | Rarely                                                                    | +/-             |
| <b>Headache</b>              | Rarely                                                                    | (+)             |
| <b>Urinary incontinence</b>  | (+)                                                                       | (+)             |
| <b>Loss of consciousness</b> | Gradual                                                                   | Sudden          |
| <b>Recovery</b>              | Second to mins (leg raise)                                                | Minutes - hours |

- **Serum prolactin levels** are always elevated after an episode GTCS

## EPILEPSY SYNDROMES

01:17:54

### • LENNOX-GASTAUT SYNDROME

- GTCS
- **Atypical** absence seizures
- Atonic seizures
- Etiology: multifactorial
- EEG: **< 3 Hz spike and wave pattern**
- T/t: **Valproate (DOC)**
- S/E: hyperammonemia